SCOPE · 3D — Graphics & Avatar Overhaul Plan

I researched mobile-friendly Three.js best practices before writing this. The mobile budget is the constraint that drives almost every decision below — your iPhone in the Instagram in-app browser is already pushing 163 city building meshes + 12 humanoids + scope post-effects + an embedded GLB rifle. We can absolutely make it look way better, but we have to be careful not to crash it.

What's getting upgraded

1. **Sky & atmosphere** — real sky shader with sun, soft horizon glow
2. **Lighting** — proper warm sun + cool sky-fill + ambient, contact shadows under characters (not real shadows — too expensive)
3. **Tone mapping & color grade** — ACES filmic + slight warmth/contrast for the cinematic look
4. **Buildings** — varied roof details (water tanks, antennas, AC units), better window emission, accent strips
5. **NPC bodies** — proportions and silhouettes that read like people from a distance, not stacks of boxes
6. **Player avatar** — same upgrade as NPCs
7. **First-person view** — visible **arms holding the rifle** so it doesn't look like a floating gun
8. **Fire button** — moves to vertically-centered LEFT (round shutter) only when scoping; back to bottom-right unscoped
9. **Performance safety net** — settings menu's "LOW" graphics mode disables the heaviest stuff so older phones still run

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Why these choices (the research)

- **Mobile lighting cap is ~3 lights, 0–1 with shadows.** Real shadow maps from a directional light over a 200m city would need cascade shadow maps (CSM) to look acceptable, and CSM doubles+ the GPU cost. Verdict: **no real-time shadows**. Use a fake “contact shadow” disc under each character + bake the city’s directional shading into the building materials’ face normals.

- **Three.js Sky shader (Preetham model)** is built-in and gives you sun, atmospheric scattering, horizon — all on the GPU at almost no fragment cost since it’s just a sky dome. This is the standard answer.
- **ACESFilmicToneMapping** is the cheapest single change that makes a Three.js scene look “next-gen”. One line. It compresses HDR-ish lighting into nicer film-like response curves so colors don’t blow out.
- **MeshLambertMaterial stays for buildings** — it’s matte and cheap. Switching them all to MeshStandardMaterial would be prettier but costs ~2× the fragment shader time × 163 buildings. Verdict: keep Lambert for buildings, upgrade to Standard only for hero objects (rifle, avatar, contracts).
- **For the humanoids**: I’m not adding skeletal rigs (would need a proper rigged GLB which we couldn’t find under 5 MB). The plan is to make the box-stack humanoids look way better through silhouette work — tapered torso, narrower waist, properly proportioned head, defined shoulder caps, varied clothing colors. Same skeleton-of-cubes architecture, much better proportions and colors. Walking animation is already there (legs/arms swing) and stays.
- **Contact shadows** are a semi-transparent dark disc on the ground under each character. Free. Looks 80% as good as real shadows for this kind of game.

What it will look like

Before (current):

- Flat sky gradient (CSS background)
- Sun/light is just a directional vector; no visible sun
- Everything feels lit from the same yellow-ish ambient
- Buildings all the same shape (box with windows)
- Humanoids: torso = single 0.95×0.75×0.42 box, head = sphere, arms hang straight, legs are dark cylinders. They look “boxy” up close.
- 1st person: floating rifle in lower-right, no hands

After:

- Real sky dome with a sun disc you can SEE, soft horizon glow, atmospheric perspective so distant buildings fade into the sky color naturally
- Sun is warm directional light (3000K-ish), sky is cool blue fill, deep ground shadow color → buildings have proper "lit side / shadow side" reading
- Tone-mapped image looks like a photo, not a flat render
- Roughly 1 in 4 buildings has a rooftop detail (water tank, antenna cluster, AC unit, billboard) so the skyline isn't all flat tops
- Humanoids have visible shoulders, narrow waist, proper hip/leg proportions — silhouette reads as "person" not "stack of boxes"
- 1st person: gloved hands gripping the rifle (left hand on forestock, right hand on grip)
- Fire button when scoped: round shutter button on the vertically-centered left edge

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Performance budget (what I'm protecting)

Going to keep these targets so it stays smooth on your phone:

Resource	Current	After upgrade	Limit
Lights	3	3	3 max
Shadow casters	0	0 (fake contact shadows)	0
Draw calls	~300	~330	<500
Triangles	~110k	~140k	<250k
Materials	~30	~40	<60
Pixel ratio (med)	1.5	1.5	1.5

Execution order (so you can stop me at any point if it breaks)

I'll do the upgrade in **5 small commits** instead of one giant rewrite. After each one I'll test on the desktop emulator and push a version. If something breaks the look on your phone you tell me which step to roll back.

Step 1 — Tone mapping + sky (biggest visual gain, smallest code change)

- Switch renderer to ACESFilmicToneMapping with exposure ~1.1
- Set color space sRGB throughout
- Add the Three.js Sky object (Preetham model) sized 6000m as the skybox
- Sun direction matched to existing sunLight position
- Fog color matched to horizon so distant buildings fade naturally
- **Risk: low.** Estimated time: small.

Step 2 — Lighting tune

- Sun light: warm 0xfff1d8, intensity 1.4, position recomputed for the new sky's sun
- Hemisphere light: sky tint 0xa8c8ff, ground tint 0x4a3a28, intensity 0.5
- Drop ambient to 0.2 (it was hiding the directional contrast)
- Add contact-shadow plane under each humanoid (a flat dark disc with radial gradient texture, semi-transparent, additively blended into the floor)
- **Risk: low.** Visual jump is significant.

Step 3 — Avatar & NPC silhouette pass

- Re-proportion the makeHumanoid: narrower waist, defined shoulders, slimmer arms, longer legs, smaller head (current head is too big)
- Cleaner color palette per archetype (soldier = olive + black tac vest, civilian = brown jacket + jeans, militia = dusty red + sand pants)
- Subtle 2-tone shading via vertex colors on the torso (top slightly lighter, bottom darker — gives "lit from above" feel even without dynamic shadows)
- Same skeleton structure — walking animation continues to work unchanged
- **Risk: medium** — touches the makeHumanoid function which everything calls. I'll keep the existing limb names so the walk-cycle code keeps working.

Step 4 — Building variety

- Add a rooftop detail builder (water tank cylinder / antenna mast / HVAC box / roof billboard)
- ~25% of buildings get one

- Window material gets a slight emission so they actually glow at dusk
- Accent stripe (a thin colored band) on every 4th building near the top
- **Risk: low** — additive only, nothing changes about existing geometry.

Step 5 — First-person hands + fire button reposition

- Build a small "left arm" group (gloved forearm + hand) attached to rifle's forestock position
- Build a "right arm" group (gloved forearm + hand) attached to grip
- Arms move with the rifle (parented to the rifle group, so reload anim makes them move correctly)
- Fire button CSS: move scoped position to round-shutter-left-center
- **Risk: medium** — hand positions need per-rifle tuning since each rifle has different geometry.

What I'm NOT doing

- **No real-time shadows.** Mobile can't afford it for a 200m city scene. Fake contact shadows only.
- **No HDR environment maps.** The .hdr file would be 2-5 MB more, and tone mapping with the sky shader gets us most of the way there.
- **No post-processing pipeline (EffectComposer).** Each post-pass is a full-screen draw at the rendered resolution. Adds noticeable cost on iPhone. Tone mapping is built into the renderer, no composer needed.
- **No skeletal animation (skinned mesh).** Would need a rigged GLB. The current "rotate the limb groups" animation is already smooth and free.
- **No bloom / depth-of-field.** Cool but expensive and we already have a CSS blur for scope transition.

After the build

I'll push v13 (tone map + sky), v14 (lighting), v15 (avatars), v16 (buildings), v17 (hands + fire button). Each is a working game so you can stop wherever it looks best on your phone.

If everything works, the result should be a clear visual upgrade — not “AAA mobile game” level (we’d need real models for that), but no longer “obviously procedural cubes”. The biggest single jump will be Step 1 (tone mapping + sky), which you’ll see immediately.